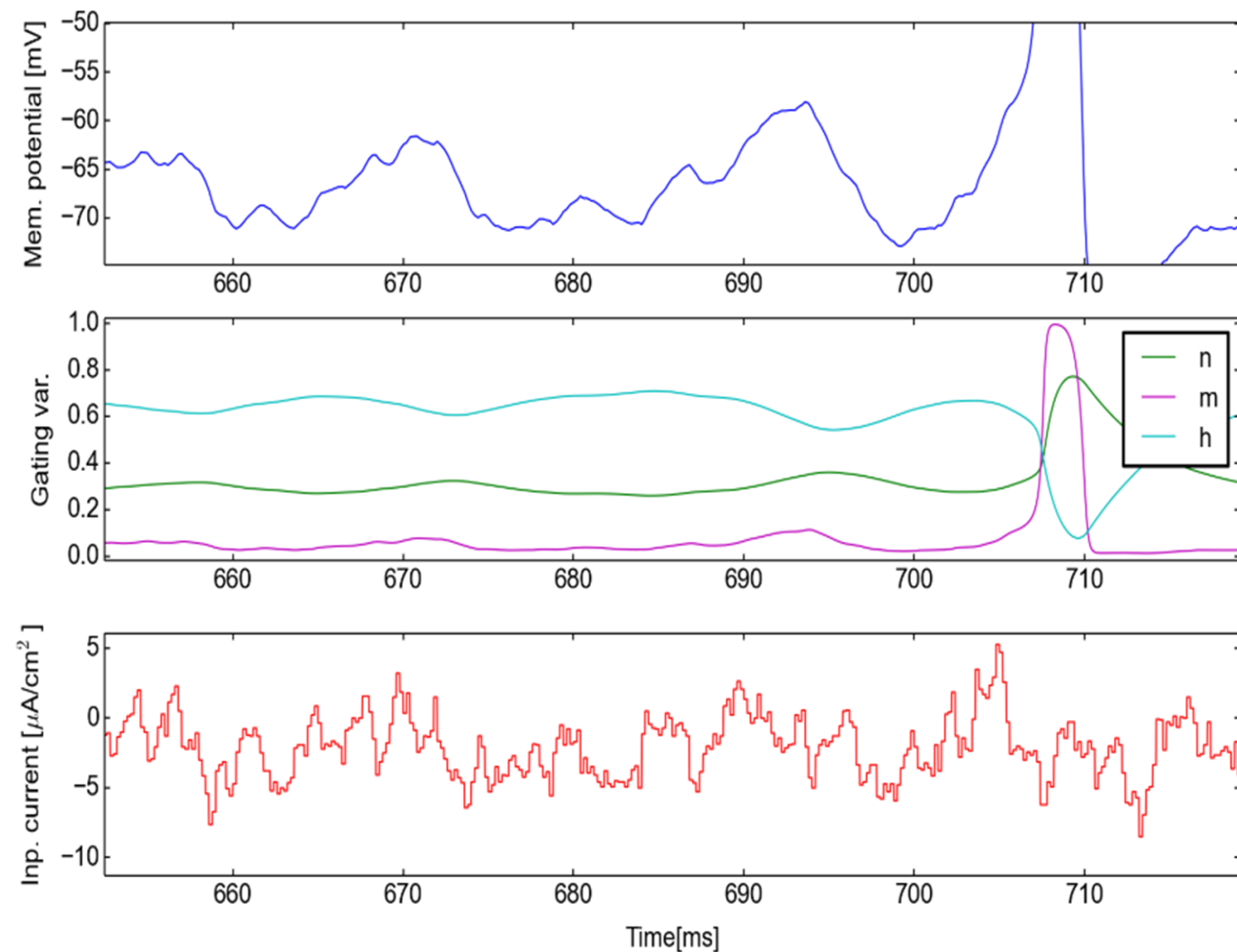
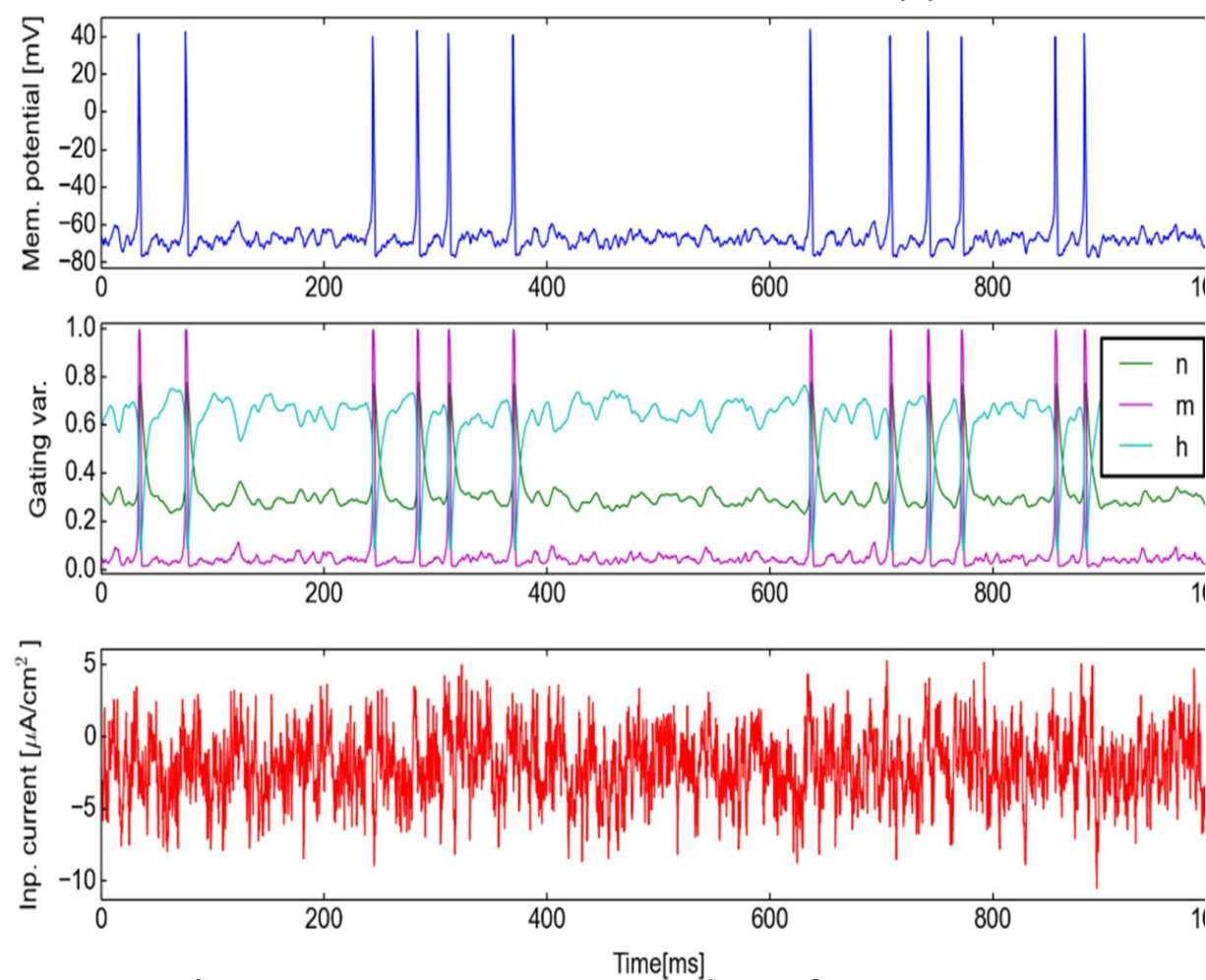


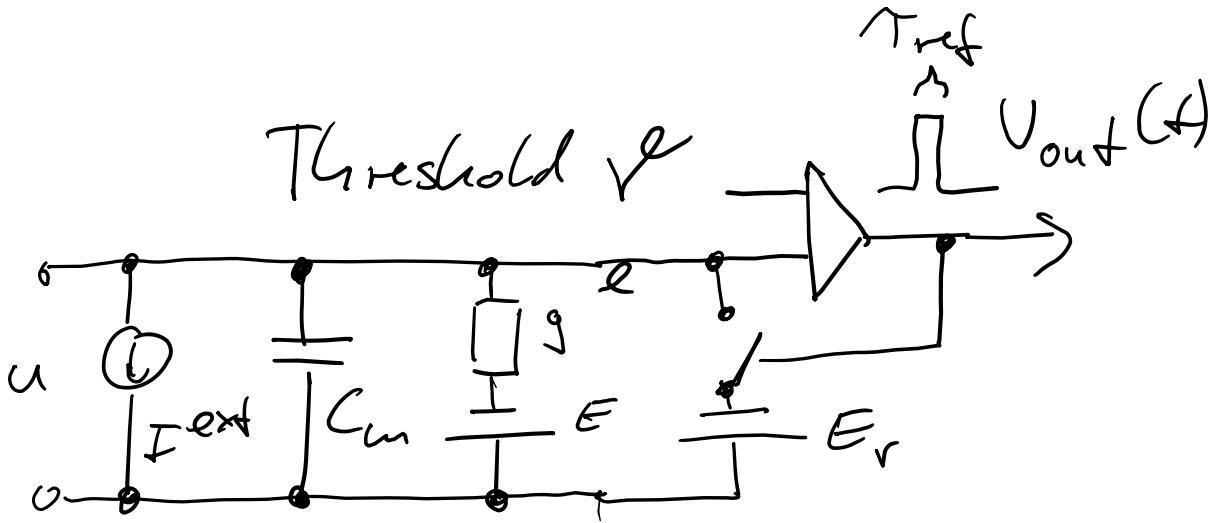
A (naive) 1-dimensional neuron model

- constant and step currents are somewhat artificial
- let's look at a more realistic stimulus $I(t)$:



- we observe two primary modes of operation:
 - sub-“threshold” integration $\rightarrow$ gating variables remain approx. constant
 - action potentials $\rightarrow$ gating variables follow stereotypical trajectories

Equivalent circuit



Leaky integrate and fire (LIF) properties

- membrane time constant $\tau_m = \frac{C_m}{g_l}$
- reset potential E_r (often $E_r = E_l$)
- absolute refractory period $\tau_{ref} \leftrightarrow$ no relative refractory period
- threshold potential ϑ
- output spike train
 - not part of membrane voltage
 - input and output are electrically separated
 - information flows always from the membrane to the spike

Leaky integrate-and-fire model (LIF):

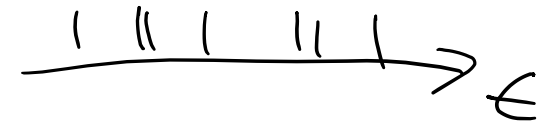
$$C_m \dot{u} = -g(u - E) + I^{ext}$$

$$u(t_s) \geq \vartheta \leftrightarrow \text{spike at } t = t_s$$

$$V_{out}(t) = \sum_s \Theta(t - t_s) \Theta(t_s + \tau_{ref} - t)$$

$$u(t_s < t < t_s + \tau_{ref}) = E_r$$

$$g(t) := \sum_s \delta(t - t_s)$$



Activation function of LIF neurons

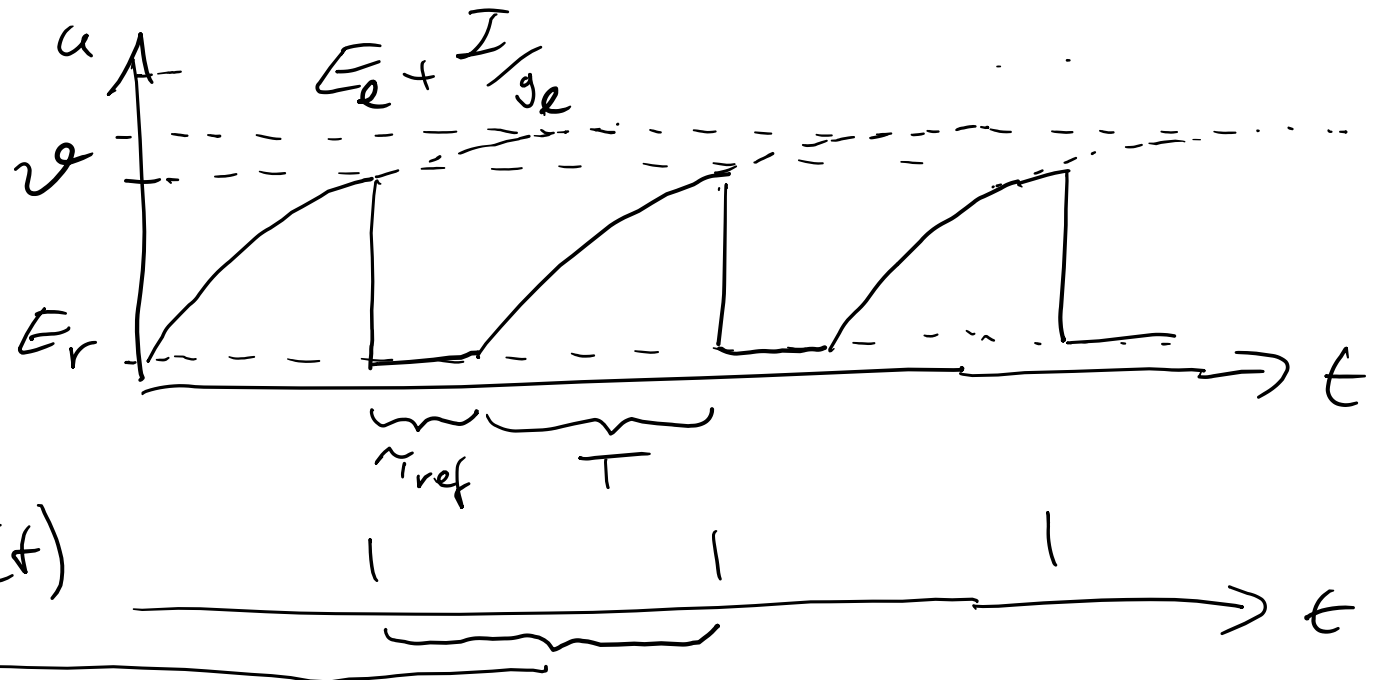
- $I^{\text{ext}}(t) = I \cdot \Theta(t)$

- Sub-threshold regime (no spikes)

Set $i=0 \rightarrow u \rightarrow E_r + \frac{I}{g_L} < v$ with $C_{ui} = -g_L(u - E_r) + I$

$\hookrightarrow v(I) = 0$ for $I \leq I_{\text{ref}} = g_L \cdot (v - E_r)$

- Spiking occurs for $I > I_{\text{ref}}$



inter-spike interval

$$\tau_{\text{isi}} = \tau_{\text{ref}} + T$$

• Calculating T for $I > I_{cr}$:

$$V \stackrel{\Delta}{=} u(T)$$

known solution for LIF: $u(t) = u_0 e^{-\frac{t}{\tau_m}} + \frac{1}{\tau_m} \int_0^t e^{-\frac{t-t'}{\tau_m}} \cdot \left(E_L + \frac{I^{ext}(t')}{g_L} \right) dt'$

$$u_0 = E_r \quad t \rightarrow T$$

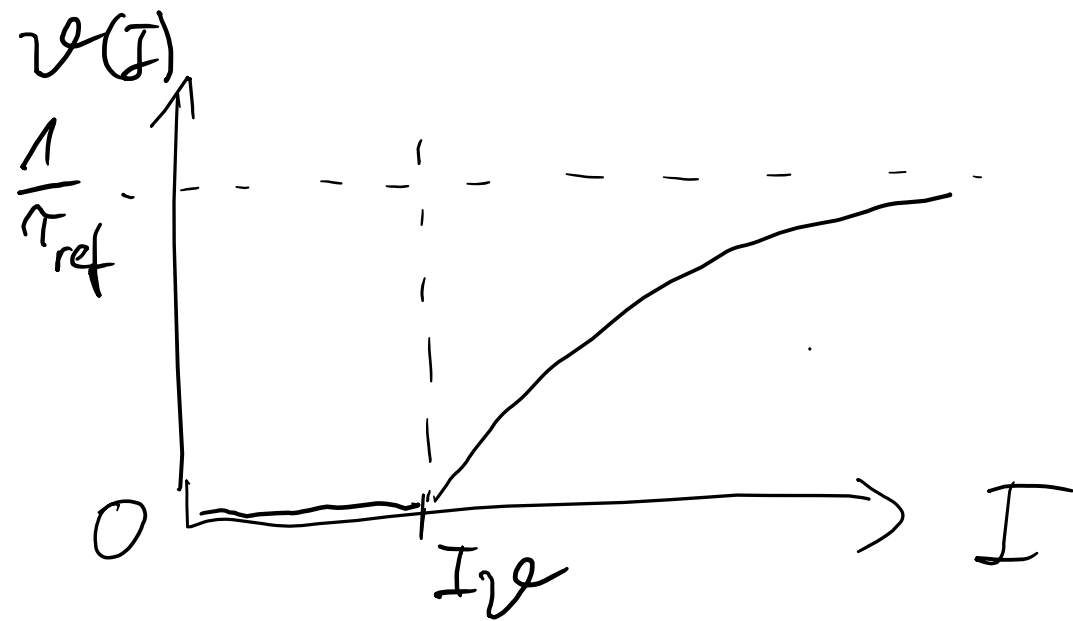
$$V = E_r e^{-\frac{T}{\tau_m}} + \frac{1}{\tau_m} e^{-\frac{T}{\tau_m}} \int_0^T \underbrace{\left(E_L + \frac{I}{g_L} \right)}_{\text{const}} \cdot \underbrace{e^{\frac{t'}{\tau_m}}}_{\int = \tau_m (e^{\frac{T}{\tau_m}} - 1)} dt'$$

$$= E_L + \frac{I}{g_L} + e^{-\frac{T}{\tau_m}} \cdot \left(E_r - \left(E_L + \frac{I}{g_L} \right) \right)$$

$$\hookrightarrow T = \tau_m \cdot \ln \left[\frac{E_L + \frac{I}{g_L} - E_r}{E_L + \frac{I}{g_L} - V} \right]$$

$$T = \tau_m \cdot \ln \left[\frac{E_l + \frac{I}{g_l} - E_r}{E_l + \frac{I}{g_l} - v} \right]$$

$$v(I) = \frac{1}{\tau_{ISI}} = \left(\tau_{ref} + \tau_m \ln \left[\frac{E_l + \frac{I}{g_l} - E_r}{E_l + \frac{I}{g_l} - v} \right] \right)^{-1}$$



summary		
model type / feature	Hodgin-Huxley	LIF
subthreshold integration	✓	✓
spiking	✓	✓
well defined ϑ	-	✓
type	2	1
inhibitory rebound	✓	-

LIF is one of the most widely used neuron models!

Extension to 2-dimensional neuron models

Simple integration of $I(t)$ is not enough.

- The HH model revisited: m gating variable is very fast (quasi instantaneous)
 - no real temporal coupling
- n gating variable controls conductance to the low E_{K+} reversal potential
 - Pulls $u(t)$ down after spike
 - Reacts „slowly“ in the inhibitory rebound scenario
→ significant role for quickly changing input
- General idea for dimensionally reduction of the HH model:
 - m : $\tau_m \rightarrow 0$, i.e. $m(t) \rightarrow m_0(u(t)) \rightarrow$ 1st ODE removed
 - n, h : $\tau_n(u) \approx \tau_h(u)$ and $n_0(u) \approx 1 - h_0(u)$
→ combine n and $1-h$ to one variable ω with time constant $\tau_\omega \rightarrow$ 2nd ODE removed

Possible approach:

Design a new (functional) model inspired by the above ideas, e.g.:

- Fitz Hugh-Nagumo (1961)
- Izhikevich: Quadratic Integrate and Fire (2003)
- Naud: Adaptive Exponential Integrate and Fire (2008)
→ Heidelberg BrainScaleS System (2010)

